Ayurvedic Plant Leaf classification using Machine Learning

Abstract

Traditional medicine has used medicinal plants for ages as all-natural treatments for a wide range of illnesses. They include medicinal bioactive substances that can be utilized to treat a variety of ailments. The identification and characterisation of medicinal plants are of increasing importance due to the rising demand for natural products and the requirement for sustainable healthcare. Based on their physical and chemical features, machine learning (ML) and deep learning (DL) algorithms have shown considerable promise for the detection and classification of therapeutic plants. Natural chemicals found in medicinal plants are a great source for the creation of novel medications and treatments.However, because there are so many diverse species with comparable physical characteristics, it can be difficult to identify and characterize therapeutic plants. Additionally, the habitat, climate, and growing circumstances all have an impact on the chemical makeup of medicinal plants. Therefore, for medicinal plants to be used effectively in medicine, correct identification and classification are essential.

When it comes to the identification and classification of medicinal plants, ML and DL approaches have shown considerable potential. These techniques can analyze big datasets and extract features that are difficult for the human eye to see. For instance, image recognition algorithms may examine photos of medicinal plants and pinpoint their distinctive morphological characteristics, such as the size, shape, and texture of their leaves.These characteristics can then be used by classification algorithms to divide medicinal plants into several groups according to their morphological or chemical characteristics.Convolutional neural networks (CNNs) for picture identification are one example of the employment of ML and DL techniques in medicinal plant detection. CNNs are DL models that can spot patterns and distinguish important details in images.CNNs have been used by researchers VIII to categorize many medicinal plants according to the shape, texture, and color of their leaves.

The detection and classification of medicinal plants based on their morphological and chemical properties has shown tremendous promise for ML and DL approaches, in conclusion. These techniques can analyze big datasets and extract features that are difficult for the human eye to see. feature extraction and image recognition.

Keywords: Medicinal plants, traditional medicine, natural remedies, bioactive compounds, sustainable healthcare, machine learning (ML), deep learning (DL), morphological, chemical characteristics, image recognition, feature extraction, classification algorithms, convolutional neural networks (CNNs), leaf morphology, leaf texture, leaf color, environmental conditions, growth conditions.

INTRODUCTION

1.1 Introduction

The use of medicinal plants has been prevalent in traditional medicine practices for centuries. These plant-based remedies have been used to treat a wide range of ailments, from minor illnesses to more serious diseases. While conventional medicines have their benefits, herbal remedies have been found to be effective, safe, and have fewer side effects. In fact, according to the World Health Organisation (WHO), up to 80% of people in underdeveloped nations use traditional medicine, with herbal medicines making up a sizable portion of this.

Despite the widespread use of medicinal plants, identifying and classifying them can be a challenging task. Traditional methods of identifying plants involve manual observation (Fig 1.1 shows random leaves images that can't be classified with the naked eye) and measurement, which can be time-consuming, subjective, and prone to errors. This is especially problematic for individuals with limited botanical knowledge, as it can lead to incorrect identification and potentially harmful consequences. To address these issues, machine learning (ML) and deep learning (DL) algorithms have been applied to classify medicinal plants based on the physical characteristics of their leaves (Fig 1.2 shows some examples of some medicinal plants leaves). This approach involves capturing digital images of plant leaves and using image processing techniques to extract relevant features such as texture, shape, and color. The extracted features are then used as inputs to ML and DLmodels to learn the patterns that distinguish different species of medicinal plants.

One of the key advantages of using ML and DL algorithms for plant classification is their ability to process large amounts of data quickly and accurately. This capability allows for the identification of many different plant species, which may not be feasible with traditional methods. Additionally, these algorithms can identify subtle differences in plant morphology that may be difficult for the human eye to detect, leading to a more accurate classification of plant species. Several studies have been conducted to validate the effectiveness of ML and DL algorithms for medicinal plant classification. For instance, a study conducted in China used a convolutional neural network (CNN) to classify seven different medicinal plants based on their leaf images. The model achieved an accuracy of 97.54%, demonstrating the potential of this approach for medicinal plant identification. Similarly, a study conducted in India used a combination of texture and shape features to classify 12 different medicinal plants. The model achieved an accuracy of 95.83%, which is comparable to the accuracy achieved by experts in traditional plant classification. Another study conducted in Brazil used a machine learning approach to classify four different species of medicinal plants. The model achieved an accuracy of 92.5%, demonstrating the potential of this approach for plant identification in regions with high biodiversity. The use of ML and DL algorithms for medicinal plant classification has several applications. For example, it can be used to detect adulteration and substitution of medicinal plants, which is a significant concern in the herbal medicine industry. By accurately identifying plant species, this approach can ensure the quality and safety of herbal medicines and prevent harmful consequences resulting from the use of incorrect plant species. Moreover, ML and DL algorithms can be used to identify new plant species with potential medicinal properties. This approach involves training models on a dataset of known medicinal plants and using them to classify unknown plants. The identified plants can then be further studied for their potential medicinal properties, leading to the discovery of new natural medicines.

In conclusion, the use of ML and DL algorithms for medicinal plant classification has the potential to revolutionize herbal medicine by improving the accuracy and efficiency of plant identification. This approach is more efficient than traditional methods, can identify subtle differences in plant morphology, and can be used to detect adulteration and substitution of medicinal plants. Moreover, it can be used to identify new plant species with potential medicinal properties, leading to the discovery of new natural medicines. As such, the future scope of using ML and DL techniques for medicinal plant detection by classifying leaves using ResNet 50 is promising. Researchers can continue to refine the models, integrate other modalities, develop portable and user-friendly applications, expand the use

Fig 1.1 : Random leaves

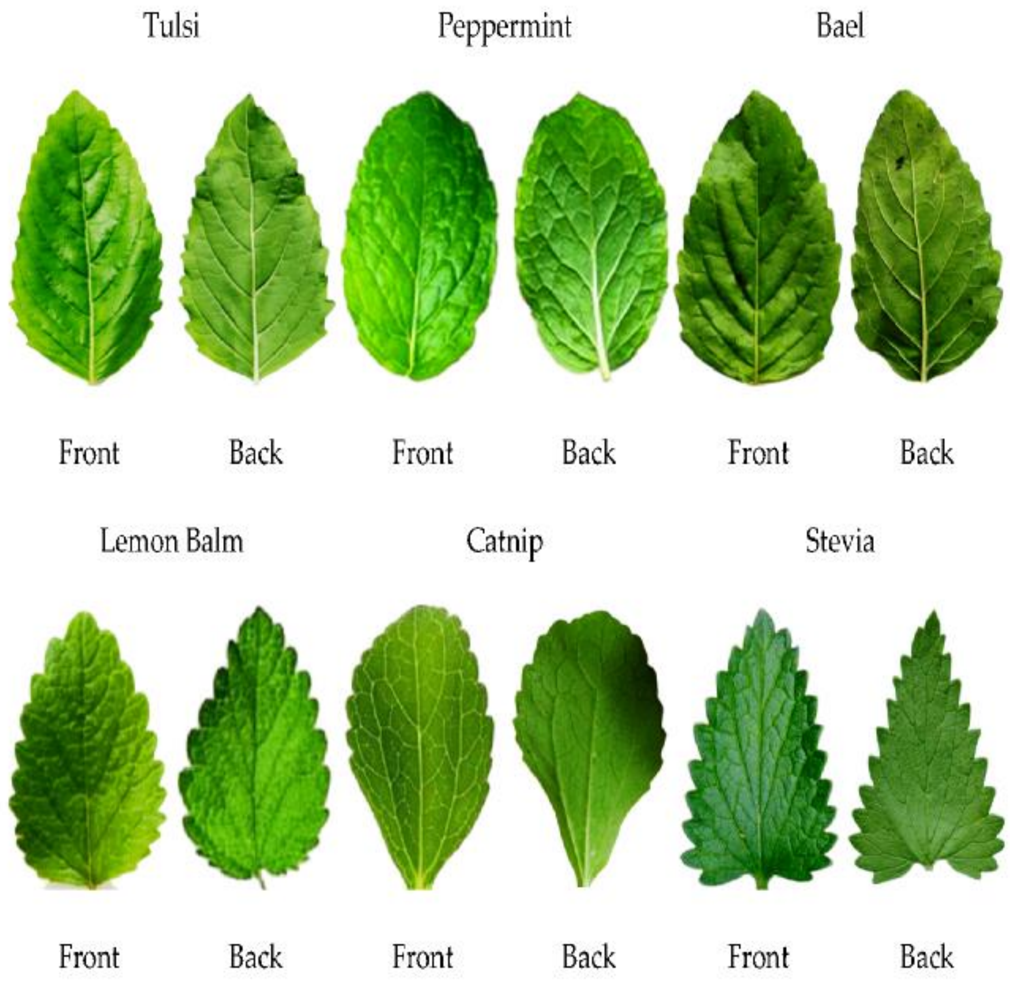

Fig 1.2 : Example of Medicinal Plants leaves

1.2 Problem Statement

The identification and classification of medicinal plants have been a challenging task for experts due to the large number of plant species, variations in plant morphology, and environmental factors. Traditional methods of identification, such as field observations and taxonomic keys, are time-consuming, expensive, and require extensive botanical knowledge. With the increasing demand for medicinal plants due to their potential health benefits and the need for sustainable and responsible harvesting practices, the development of an accurate and efficient system using Machine Learning (ML) and Deep Learning (DL) algorithms to identify and classify different types of medicinal plants based on their images is essential. This system will be developed using advanced ML and DL algorithms that can analyze visual features and patterns in plant images to identify and classify different types of medicinal plants. The system will be trained using a large dataset of images of different medicinal plant species, along with their corresponding labels. The images will be preprocessed to remove any noise and enhance the features that are important for classification. The ML and DL algorithms used in this system will be capable of recognizing complex patterns in the images, enabling accurate identification and classification of medicinal plants. The system will also be able to adapt to variations in plant morphology and environmental factors, making it more robust and accurate. The system will be developed using open-source ML and DL frameworks such as TensorFlow, PyTorch, and Keras, making it easy to customize and integrate into existing applications.

The development of this system will have a significant impact on various fields, such as traditional medicine, agriculture, and biodiversity conservation. Experts in traditional medicine will be able to easily and quickly identify medicinal plants, which is essential for their proper use in traditional medicine and drug discovery. The system will also be useful in the agricultural industry, where it can help farmers identify and classify different types of medicinal plants, enabling them to make better decisions about their cultivation and harvesting practices. Additionally, the system can aid in biodiversity conservation efforts by enabling experts to accurately identify and classify different types of medicinal plants in natural habitats.

In conclusion, the development of an accurate and efficient system using ML and DL algorithms to identify and classify different types of medicinal plants based on their images is a crucial step towards sustainable and responsible harvesting practices. This system will enable experts in traditional medicine, agriculture, and biodiversity conservation to easily and quickly identify medicinal plants, which is essential for their proper use in traditional medicine and drug discovery.

1.3 Objectives

Developing an efficient and accurate method for identifying medicinal plant species based on their leaf images is a crucial step towards sustainable harvesting practices. To achieve this goal, the collected images need to be preprocessed and data augmentation techniques should be employed to increase the size and diversity of the dataset. The dataset should contain images of various medicinal plant species, each with their corresponding label.

To develop a deep learning model for detecting medicinal plants, advanced algorithms like convolutional neural networks (CNNs) can be utilized. These models can learn complex features and patterns in the images, making them ideal for plant identification tasks. The goal is to develop an automated and efficient method for accurately identifying and classifying different plant species based on unique leaf characteristics. Once the model is developed, it needs to be evaluated using a validation set and various metrics such as accuracy, precision, recall, and F1 score. The performance of the model can be further improved by fine-tuning the hyperparameters and using transfer learning techniques. For instance, ResNet50, a pre-trained CNN model, can be used as a base model, and its weights can be fine-tuned to improve the accuracy of the medicinal plant classification task.

After training and validating the model, it can be used to classify new images of medicinal plant leaves. A user-friendly web application can be developed for this purpose, where users can easily upload images of plant leaves and get instant results. The web application can be developed using popular web frameworks like Flask or Django, along with front-end libraries like React or Vue.

In conclusion, the development of an automated and accurate method for identifying medicinal plant species based on their leaf images is an essential step towards sustainable harvesting practices. By utilizing deep learning models, pre-processing techniques, and data augmentation, we can develop a model that is robust and efficient in identifying various medicinal plant species. The use of transfer learning techniques can also improve the performance of the model. With the development of a user-friendly web application, experts in traditional medicine, agriculture, and biodiversity conservation can easily and quickly identify medicinal plants.

LITERATURESURVEY

In many applications, like plant recognition, face recognition, etc., an image conveys the most valuable information as opposed to the natural description. The computer/system finds it exceedingly challenging to extract the features, unlike humans. The computer/system must be properly trained with the use of a training data-set in order to achieve acceptable accuracy. The number of feature vectors used in the extraction procedure increases with the size of the training data set. Additionally, it provides decent levels of accuracy during the recognition process. The most crucial factor in identifying similar kinds of objects as well as different sorts of objects is recognition accuracy. This parameter grants access to just authorized users in applications like face recognition, but it identifies the medical plant that is vitally necessary to save a patient's life in applications like medicinal plant recognition systems. Ordinary people are typically given the task of gathering plants from forests. Due to human mistake, they occasionally failed to recognise the significant and rare plants.

These exotic plant species are crucial to preserving a patient's life. Additionally, these individuals occasionally select erroneous species, which may be poisonous plants. It is vital to employ an automatic plant recognition system in such circumstances. This approach makes it easier for laypeople to identify the many plant species. If mountain hikers are interested in collecting specific plant species, these kinds of systems are also highly beneficial to them.

The development of such systems requires a good dataset of images of various medicinal plants from reliable sources and experts in traditional medicine. Once the dataset is collected, it needs to be preprocessed and cleaned to remove irrelevant or duplicate images, and to enhance the quality of the images for better performance. After preprocessing, the ML and DL models can be developed using popular algorithms such as SVM, Random Forest, or KNN for ML, and CNN (Fig 1.4 shows the CNN Architecture) for DL (using ResNet 50 here). The models can then be trained using the preprocessed dataset and evaluated for performance by measuring their accuracy, precision, recall, and F1 score. The comparison of the performance of the ML and DL models can help determine which model is more suitable for this problem. Once a suitable model is selected, a web or mobile application can be developed to allow users to take pictures of medicinal plants and classify them using the trained model. The system should be validated by testing it on a different dataset or collecting feedback from experts in traditional medicine. Documentation and reporting are also important aspects of the project work, including dataset collection, preprocessing, model development, and application development. The findings and the performance of the developed system should be reported for future reference and improvements. In summary, an automatic plant recognition system can be a valuable tool for identifying rare and important medicinal plants, and can potentially save lives in critical situations.[1]

The separation of therapeutic plants from other non-edible plants is crucial in the realms of botany and the food industry. However, conventional techniques for identifying medicinal plants are difficult, time-consuming, and require skilled specialists. An autonomous real-time vision-based system has been presented to identify commonly used medicinal herbs with similar leaves in order to solve this problem. This system makes use of a convolutional and classifier block-based upgraded convolutional neural network (CNN) network. Global Average Pooling (GAP), dense, dropout, and softmax layers are all present in the classifier. This technique improves the model's speed and accuracy while reducing the number of parameters compared to earlier studies. With overall accuracy rates of 99.66%, 99.32%, and 99.45%, respectively, the proposed CNN model (Fig 1.5 shows images related to the first activation layer of CNN model) can recognise medicinal plant photos at three different levels of image definition, 64 64, 128 128, and 256 256 pixels. As a result, combining image processing with the suggested CNN algorithm is a productive replacement for conventional approaches. To verify the efficacy of the developed approach, additional work will be done to enhance the model's performance in the classification of additional species of medicinal plants. A smart smartphone application for the real-time identification of medicinal plants will also be created using the model. This is especially crucial in light of the rising acceptance and demand for both artisanal and commercial uses and applications of medicinal plants. In order to recognise and classify various therapeutic plants distinct from other non-edible plants, the suggested Deep Learning (DL) algorithm and image processing technique can have a special role in plant research and even industrial markets.[2]

Plants have played an essential role in human survival since the dawn of civilization, providing nourishment, shelter, and medicine. Among the most intriguing aspects of plant utilization is herbal medicine, a practice that has been passed down through generations of indigenous peoples. For centuries, these remedies have been identified by experienced clinicians who rely on their senses and extensive knowledge of plant properties. However, recent technological advances have provided an evidence-based approach to herb identification, making it easier for individuals who are not familiar with these practices. Various techniques are available for herb identification, each with its own advantages and limitations. One promising method is based on spectral analysis, a non-invasive technique that can distinguish plant species based on their unique spectral signatures. Hyperspectral imaging (HSI) and near-infrared spectroscopy (NIRS) are advanced instruments used in spectral analysis to capture the spectral signature of a plant. This approach is rapid and non-destructive, but it requires expensive equipment and trained personnel, which can be a barrier for many people. Another promising technique is based on computer vision and machine learning algorithms, which use artificial intelligence (AI) to analyze plant images and identify them based on their unique characteristics. Deep learning (DL) algorithms have been particularly successful in identifying herbs with high accuracy rates. However, this approach requires a large amount of high-quality training data, which can be challenging to obtain. Traditional morphological and chemical analysis can also be used for herb identification. Although these methods have been used for centuries, they can be time-consuming and require expertise in sample preparation and data interpretation. Nevertheless, they remain an important tool in the herbal identification toolkit, especially when used in combination with modern analytical techniques. In conclusion, identifying herbs is a crucial aspect of herbal medicine and natural product research. Modern analytical techniques, such as spectral analysis and computer vision (Fig 1.6 shows the flowchart of proposed system), have shown great potential in providing a rapid and reliable method for herb identification. However, their success depends on the availability of high-quality training data and specialized equipment. Therefore, a combination of these modern techniques with traditional methods may provide the most effective and comprehensive approach for herb identification. As herbal medicine gains popularity in mainstream healthcare, the need for accurate and efficient herb identification will continue to grow.[3]

A statistical analysis of leaf characteristics has been conducted to identify the key features that aid in plant identification, and form was found to be a crucial factor. This promising approach has the potential to aid individuals in identifying medicinal plants automatically, as well as in conservation and utilization efforts. The development of an artificial intelligence system for plant recognition is essential to achieving these objectives, as it can process large amounts of data efficiently and accurately. Moreover, the proposed system's (Fig 1.7 shows the steps of the proposed method) accuracy will undoubtedly improve as more data is collected and analyzed. In addition, the use of advanced image processing techniques and machine learning algorithms can aid in identifying complex features and patterns in plant images, resulting in a more reliable and efficient recognition system (Fig 1.8 shows the process of medical plant recognition).

Therefore, the application of artificial intelligence in plant recognition can provide valuable insights into the field of botany and have a significant impact on the pharmaceutical industry, as well as on conservation efforts for rare and endangered plant species. Further research and development in this area will undoubtedly yield more accurate and efficient plant recognition systems that can assist individuals in identifying medicinal plants quickly and accurately.[4]

1. Existing System

- **Description**:
  Existing systems for medicinal plant identification rely heavily on manual efforts or rudimentary image processing techniques. Botanists or agricultural experts traditionally perform plant identification using physical inspection and taxonomical guides. In the digital space, some systems utilize basic image recognition techniques but are constrained by predefined rules and limited datasets.
- **Technologies Used**:
  - Manual classification using botanical knowledge.
  - Basic feature extraction methods like texture analysis or color histograms.
  - Limited adoption of traditional machine learning methods such as SVM (Support Vector Machines).

2. Disadvantages of the Existing System

- **Inaccuracy**: Traditional systems often fail to distinguish between plants with similar leaf structures, leading to misidentification.
- **Time-Consuming**: Manual methods are labor-intensive and slow, particularly when dealing with large datasets or field samples.
- **Limited Scalability**: Systems are often trained on small, localized datasets, making them unsuitable for diverse geographical regions.
- **Environmental Sensitivity**: Variations in lighting, background noise, and camera quality can severely affect detection accuracy.
- **Dependency on Experts**: Manual identification requires trained botanists, making the process less accessible for non-experts.

3. Proposed System

- **Objective**:
  Develop a robust machine learning-based system to detect and classify medicinal plant leaves accurately, ensuring scalability and usability in diverse environments.

- **Features**:
  - **Deep Learning Models**: Use Convolutional Neural Networks (CNNs) for feature extraction and classification.
  - **Large Dataset Training**: Train the model on an extensive dataset to improve accuracy and generalization.
  - **Environment-Agnostic**: Implement pre-processing techniques to handle variations in lighting and background.
  - **Mobile Accessibility**: Develop a user-friendly mobile application for farmers, researchers, and enthusiasts.
  - **Feedback Mechanism**: Allow users to report misclassifications for continual system improvement.

4. Requirements

A. Functional Requirements

- Detect and classify leaves from an image input into predefined categories of medicinal plants.
- Provide detailed information about identified plants, including medicinal properties and applications.
- Enable offline functionality for use in remote areas.

B. Non-Functional Requirements

- **Accuracy**: Target over 95% classification accuracy across diverse datasets.
- **Performance**: Real-time detection with minimal lag, ensuring results within seconds.
- **Scalability**: Ability to expand the system with new plant species and datasets.

- **Usability**: Intuitive interface for users with no technical background.

C. Hardware Requirements

- Camera-enabled device for capturing leaf images (e.g., smartphone or tablet).
- Computing device with GPU for model training (e.g., NVIDIA Jetson, high-performance PC).

D. Software Requirements

- **Programming Language**: Python for core development.
- **Machine Learning Frameworks**: TensorFlow or PyTorch for deep learning.
- **Image Processing**: OpenCV for pre-processing leaf images.
- **Dataset Annotation**: Tools like LabelImg for dataset preparation.
- **Platform Compatibility**: Android/iOS for mobile app deployment; web application for broader accessibility.

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